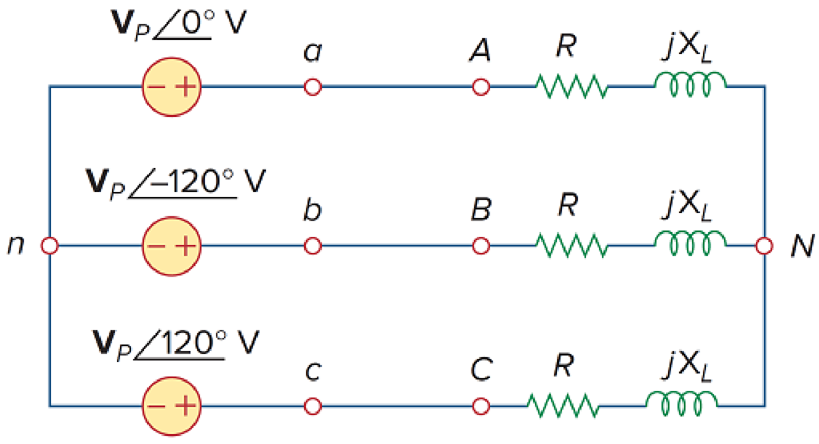
Chapter 12, Problem 6P

Problem

Using Fig. 12.41, design a problem to help other students better understand balanced wye-wye connected circuits.

Figure 12.41



Step-by-step solution

Step 1 of 1

Using Fig. 12.41, design a problem to help other students better understand balanced wye-wye connected circuits.

Although there are many ways to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

For the Y-Y circuit of Fig. 12.41, find the line currents, the line\ voltages, and the load voltages.

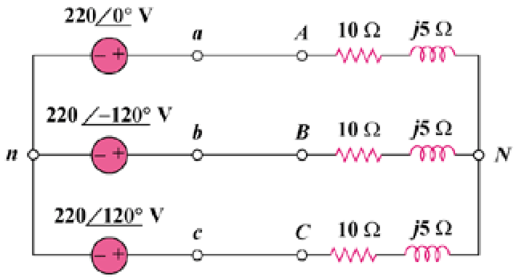


Figure 12.41

$$\mathbf{Z}_Y = 10 + j5 = 11.18\angle 26.56°$$

The line currents are

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} = \frac{220\angle 0°}{11.18\angle 26.56°} =$$

$$\mathbf{19.68\angle -26.56° \ A}$$

$$\mathbf{I}_b = \mathbf{I}_a\angle -120° =$$

$$\mathbf{19.68\angle -146.56° \ A}$$

$$\mathbf{I}_c = \mathbf{I}_a\angle 120° =$$

$$\mathbf{19.68\angle 93.44° \ A}$$

The line voltages are

$$\mathbf{V}_{ab} = 220\sqrt{3}\angle 30° =$$

$$\mathbf{381\angle 30° \ V}$$

$$\mathbf{V}_{bc} =$$

$$\mathbf{381\angle -90° \ V}$$

$$\mathbf{V}_{ca} =$$

$$\mathbf{381\angle -210° \ V}$$

The load voltages are

$$\mathbf{V}_{AN} = \mathbf{I}_a\mathbf{Z}_Y = \mathbf{V}_{an} =$$

$$\mathbf{220\angle 0° \ V}$$

$$\mathbf{V}_{BN} = \mathbf{V}_{bn} =$$

$$\mathbf{220\angle -120° \ V}$$

$$\mathbf{V}_{CN} = \mathbf{V}_{cn} =$$

$$\mathbf{220\angle 120° \ V}$$